

Quiz 10

Instructions: There are totally 2 problem. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Determine the appropriate form for a particular solution y_p of the nonhomogeneous differential equation:

$$y'' - 2y' - 3y = xe^{-x}.$$

Solution: Characteristic equation

$$r^2 - 2r - 3 = 0$$

$$(r-3)(r+1) = 0$$

$$r_1 = 3, r_2 = -1$$

$$\Rightarrow y_c(x) = c_1 e^{3x} + c_2 e^{-x}$$

Thus, the particular solution $y_p(x)$ has a form of

$$y_p(x) = x^s (Ax+B)e^{-x} \quad \leftarrow f(x) = xe^{-x}$$

$$\stackrel{s=1}{=} (Ax^2+Bx)e^{-x}$$

Problem 2. (5 points) Find the Laplace transforms of the first two functions and the inverse Laplace transforms of the last two functions:

(a) $f(t) = -2e^{3t}$

$$\mathcal{L}\{-2e^{3t}\} = -2\mathcal{L}\{e^{3t}\}$$

$$= -\frac{2}{s-3}, \quad s > 3$$

(b) $f(t) = \sin 3t + 2 \cos 3t$

$$\mathcal{L}\{\sin 3t + 2 \cos 3t\}$$

$$= \mathcal{L}\{\sin 3t\} + 2\mathcal{L}\{\cos 3t\}$$

$$= \frac{3}{s^2+9} + \frac{2s}{s^2+9}$$

$$= \frac{2s+3}{s^2+9}$$

(c) $F(s) = \frac{6}{s^4}$

$$\mathcal{L}^{-1}\left\{\frac{6}{s^4}\right\} = t^3$$

(d) $F(s) = \frac{4}{s^2-4}$

$$\mathcal{L}^{-1}\left\{\frac{4}{s^2-4}\right\} = \mathcal{L}^{-1}\left\{\frac{2 \cdot 2}{s^2-2^2}\right\}$$

$$= 2\mathcal{L}^{-1}\left\{\frac{2}{s^2-2^2}\right\}$$

$$= 2 \sinh(2t)$$